

A 276 nW, Area-Efficient CMOS Subbandgap Reference Circuit

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Abstract—An ultra low power CMOS voltage reference has been developed in a low-cost CMOS 130 nm technology. The design benefits in particular from using only one standard CMOS process without incorporating any resistors, op-amp, bipolar transistor or other CMOS processes. The circuit draws ultra low current of 230 nA from supply voltage of 1.2 V. The proposed circuit generates output an voltage of 698.5 mV with a temperature coefficient (TC) of 107 ppm/°C in a range from 0 °C to 120 °C. It achieves superior coefficient of variation σ/μ , of 0.79 % respect to other reported state-of-the-art solutions. It also features a very small size of 78.2 $\mu\text{m} \times 91 \mu\text{m}$ with an active area of 0.007 mm².

I. INTRODUCTION

The ultra low-power designs are becoming increasingly essential due to popularity and expansion of next generation application in the area of life-assistant medical devices, internet of things (IOT) or passive RFID, where they must operate for a long time under power-limited conditions [1]. In order to achieve ultra low power consumption, usually circuits employ subthreshold operating region of MOSEFT or operating with lower supply voltage. However, operating the circuits with low supply voltage usually requires additional DC-to-DC converter to make these designs compatible with existing power sources which leads to energy loss due to limited conversion efficiencies and employing subthreshold region demands for careful design to address the sensitivity of design parameters to temperature and process variations.

As a step towards such an ultra low-power system design, we should first develop voltage reference circuits as they are used almost in all analog and digital systems to generate a DC voltage independent of temperature, process, and supply voltage variations. Conventionally, bandgap voltage references with CMOS-based parasitic vertical BJTs are used for this purpose [2]. However, bandgap circuits generate a reference voltage at 1.25 V requiring a higher supply voltage and it usually employs an operational amplifier to generate proportional-to-absolute-temperature (PTAT) voltage that mostly exceeds the power-budget restriction of nanowatt range [3]. Most of conventional reference circuits also incorporate one or more resistors. In order to achieve ultra low current, the value of these resistors need to be very large within the range of several megaohm which will increase the chip area significantly, and make the design sensitive to process variation of the resistors values [2]. The other technique employs CMOS with different threshold voltage, but this approach requires further fabrication steps and also the threshold voltage of CMOS is a process dependent parameter [4].

To solve these problems, we are limiting our design to only standard CMOS process and avoid using parasitic BJTs or

other CMOS processes. We achieve ultra low power consumption by benefiting most from operating in subthreshold region. The reference voltage is generated by adding two voltages with opposite temperature coefficients, namely PTAT and CTAT. To generate the PTAT voltage, we are leveraging from a source-coupled pair used in [5] and [6] that eliminate demands for operational amplifier to provide the PTAT voltage. In order to generate the CTAT voltage, we are using a self-cascaded diode which has higher output resistance compared to a standard cmos diode with the same aspect ratio as a substitute to BJTs transistors. The design also avoids incorporating any resistor in order to minimize the size. Measurements of the fabricated chip confirm that the proposed design provides ultra low power consumption comparable to best solutions found in the literature. Among the reported nanowatt range circuits, the proposed circuit features the smallest chip area together with the minimum coefficient of variation.

II. OPERATING PRINCIPLE

The proposed voltage reference generator is shown in Fig. 1. It consists of a startup circuit, a current reference circuit, a self-cascode CMOS diode and PTAT generators. All transistors are operating in subthreshold region except M_{R1} in the current source subcircuit that works in linear region as well as M_1 and M_2 in the self-cascode diode operate in linear and saturation regions, respectively.

By considering an nMOS diode as a substitute of an BJT transistor to generate a complementary-to-absolute-temperature (CTAT) voltage, and assuming that the nMOS diode is operating in saturation region, the following equation is valid for the gate-to-source voltage of the nMOS diode:

$$V_{GS} = V_{TH} + \sqrt{\frac{2I}{\frac{W}{L}\mu C_{OX}}} \quad (1)$$

The threshold voltage can be approximated by a linear function:

$$V_{TH} = V_{TH0} - \kappa T \quad (2)$$

, where κ is the TC of V_{TH} . It shows that the current has to be proportional to μT^2 to eliminate temperature dependence of the threshold voltage. Moreover, in order to fully compensate the negative slope of the threshold voltage either the current has to be large or an additional PTAT voltage has to be added to the V_{GS} . In order to achieve minimum power consumption, the latter mentioned solution is preferred.

In our design, we benefit from the PTAT voltage generator shown in Fig. 2 where all transistors work in the subthreshold region. When $V_{DS} > 0.1 \text{ V}$, the voltage V_{GS} of transistor in

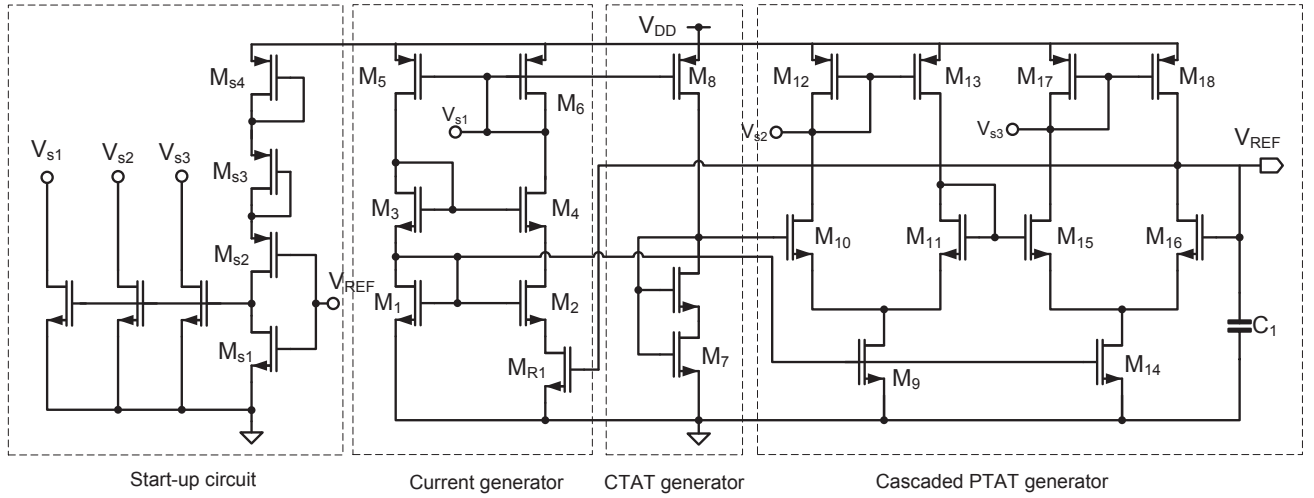


Fig. 1: Proposed reference voltage generator circuit.

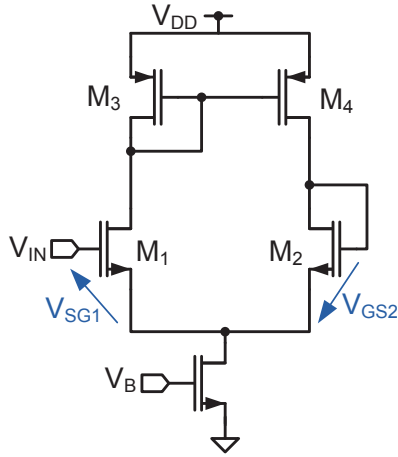


Fig. 2: Principle of the PTAT voltage generator.

subthreshold can be approximated by:

$$V_{GS} = V_{TH} + \eta V_T \ln \left(\frac{I_D}{K I_0} \right) \quad (3)$$

, where K is the aspect ratio of transistor, η is the subthreshold slope factor, and I_0 is the process-dependent parameter. The voltage difference between the gates of M_2 and M_1 is a PTAT voltage corresponding to the following equations:

$$\begin{aligned} \Delta V_G &= V_{GS2} - V_{GS1} \\ \Delta V_G &= \eta V_T \ln \left(\frac{W_1 W_4 L_2 L_3}{L_1 L_4 W_2 W_3} \right) \end{aligned} \quad (4)$$

Equation (4) suggests that ΔV_G increases linearly with temperature and in order to increase the PTAT voltage, the ratio of $\frac{W_1}{L_1}$ should be much larger than $\frac{W_2}{L_2}$ and M_4 should also draw more current than M_3 .

After indicating a technique to generate CTAT and PTAT voltages, in the next step, a current source has to be developed with characteristics of $I \propto \mu T^2$. The schematic of the current

source subcircuit can be seen in Fig. 1. The current source consists of the transistors $M_1 - M_6$ and M_{R1} . The pMOS current mirrors are sized equally to ensure both branches draw the same current. M_1 and M_2 operate in the subthreshold region and are sized differently; therefore, the resulted gate-source voltage difference drops on M_{R1} .

$$V_{DS, M_{R1}} = V_{GS1} - V_{GS2} = \eta V_T \ln \left(\frac{L_1 W_2}{W_1 L_2} \right) \quad (5)$$

Assuming M_{R1} is biased in the deep triode region then the generated current can be written as:

$$I = K_{M_{R1}} \mu C_{OX} (V_{GS, M_{R1}} - V_{TH, M_{R1}}) V_{DS, M_{R1}} \quad (6)$$

The voltage difference of $(V_{GS, M_{R1}} - V_{TH, M_{R1}})$ needs to increase linearly with temperature in order to make the generated current proportional to μT^2 . To fulfill this condition and guarantee proper operating region of M_{R1} , the gate of M_{R1} is biased by the generated reference voltage which results in a linearly increasing voltage difference of $(V_{GS, M_{R1}} - V_{TH, M_{R1}})$ since $V_{GS, M_{R1}} = V_{REF}$ is ideally constant over temperature. The startup subcircuit consisting of $M_{s1} - M_{s4}$ is used to ensure the stable state in the designed bias point which ideally has no extra power consumption when the circuit is in the steady state condition.

Ultimately, the reference voltage is generated by feeding the specified current generated by the current source subcircuit into the composed diode CTAT load and summing the resulted voltage with two cascaded PTAT stages. Consequently, resulted output voltage can be formulated as the sum of the gate-source voltage of M_7 and the PTAT voltage generated by $M_9 - M_{18}$.

$$V_{REF} = V_{GS7} + \eta V_T \ln \left(\frac{W_{10} W_{13} L_{11} L_{12}}{L_{10} L_{13} W_{11} W_{12}} \right) \quad (7)$$

The sizing of both PTAT voltage stages are identical. The transistors of the current mirrors (M_{12} , M_{13} , M_{17} , and M_{18}) are sized in a way that the currents flowing through M_{11} and M_{16} are much greater than the currents flowing through M_{10} and M_{15} . Furthermore, the aspect ratios of M_{11} and M_{16} are sized smaller to increase the PTAT voltage further. The

generated PTAT voltage is almost independent of the current and is insensitive to process variations, allowing for biasing the source-coupled pair with a very low current. Two cascaded stages of PTAT have been used, since one stage was unable to produce a sufficiently high PTAT voltage to compensate the temperature coefficient of the gate-source voltage of M_7 . The resulting reference voltage equals the threshold voltage of the used CMOS at 0 K .

$$V_{REF} = V_{TH0} \quad (8)$$

III. SIMULATION AND MEASUREMENT RESULTS

The proposed voltage reference circuit has been implemented in 130 nm standard CMOS technology. The die photograph and its corresponding layout is shown in Fig. 3. The circuit features smaller size with respect to comparable works [5] [7]. The circuit draws an 230 nA from a 1.2 V

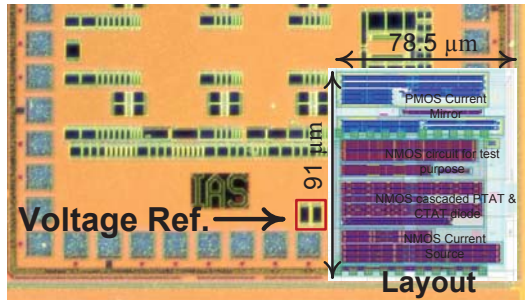


Fig. 3: Die photograph of proposed voltage reference and layout view of the circuit.

supply voltage at room temperature, which each current source branches and the CTAT subcircuit draw 60 nA and each PTAT subcircuit draw 25 nA from supply that results in total power consumption of 276 nW .

The post-layout simulation of power supply rejection is shown

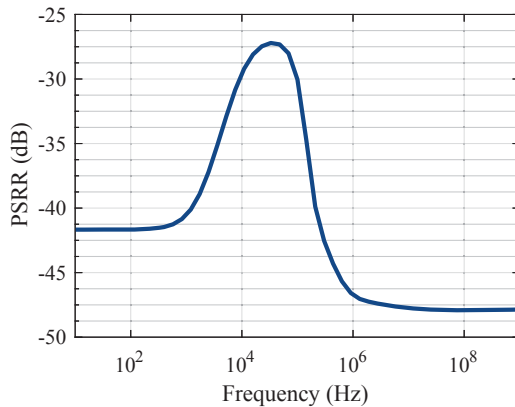


Fig. 4: Simulated PSRR at room temperature for supply voltage of 1.2 V .

in Fig. 4. The circuit achieves a PSRR of -41.5 dB at 400 Hz , in the worst case it has -27 dB at 5 KHz , and for high frequencies the PSRR is as good as at -47 dB .

On-wafer measurements of the chip have been performed using

DC-prober equipment with a thermochuck ERS Wafertherm TA2C6 and Keithely (DC voltmeter). The measurement of the sample with average performance is presented and compared with best solutions reported; then statical analysis of 12 measured samples are provided.

Fig. 5 shows the measured output voltage as a function of

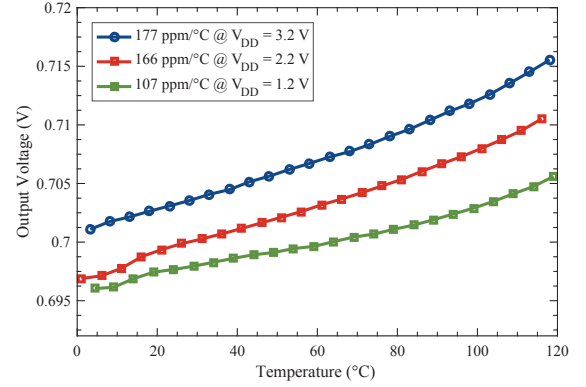


Fig. 5: Measured output voltage dependence on temperature for different supply voltages.

temperature from 0°C to 120°C for different values of supply voltage. The measured temperature coefficient at 1.2 V , 2.2 V , and 3.2 V are $107\text{ ppm}/^\circ\text{C}$, $166\text{ ppm}/^\circ\text{C}$, and $177\text{ ppm}/^\circ\text{C}$ respectively. This shows quite a stable temperature coefficient to supply voltage variations corresponding to 9 mV to 14 mV variation in the span of 120°C temperature change.

In Fig. 6, the voltage reference measured at room temperature

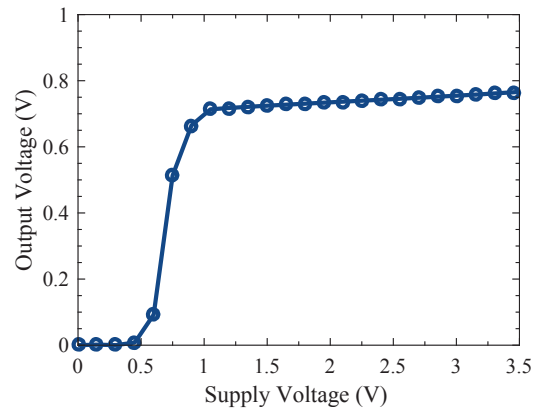


Fig. 6: Measured output voltage as a function of supply voltages at room temperature.

as a function of supply voltage is reported. The circuit starts working properly for supply voltages larger than 1 V . The mean output voltage is 740 mV for a supply voltage in the range of 1 V to 3.5 V while the line sensitivity is $2.8\%/V$.

Fig. 7 shows the measured output voltage from 12 samples as a function of temperature from 0°C to 120°C under a 1.2 V supply voltage. The average TC of the output voltage is $132\text{ ppm}/^\circ\text{C}$. Fig. 8 shows the distribution of the output voltage in 12 samples at room temperature. It results in a mean value, μ , and a standard deviation, σ , of 698.5 mV

TABLE I: Comparison with state of the art low power voltage reference

	This work	[5]	[7]	[3]	[8]	[9]	[4]
CMOS Technology	130 nm	180 nm	350 nm	350 nm	160 nm	350 nm	180 nm
Temperature range ($^{\circ}C$)	0 - 120	-40 - 120	-20 - 80	-50 - 150	-45 - 135	0 - 80	0 - 125
Supply voltage (V)	1 - 3.5	0.7 - 1.8	1.4 - 3	1.7 - 3.5	1.1 - 1.8	0.9 - 4.0	0.45 - 2.0
Reference voltage (mV)	698.5	548	745	618	944	670	263.5
Power dissipation (μW)	0.276	0.0525	0.3	65.2	1.54	0.036	0.007@0.45V
PSRR (dB)	-41.5 (@100Hz)*	-56 (@100Hz)	-45 (@100Hz)	N/A	N/A	-47 (@100Hz)	-45 (@100Hz)
TC (ppm/ $^{\circ}C$)	107	114	7	13.7	30	10	142
Coefficient of variation σ/μ	0.79 %	1.05 %	0.87 %	N/A	0.8 %	N/A	61 %
Active area (mm ²)	0.007	0.0246	0.055	0.12	0.0025	0.045	0.043

*Extracted from simulation result.

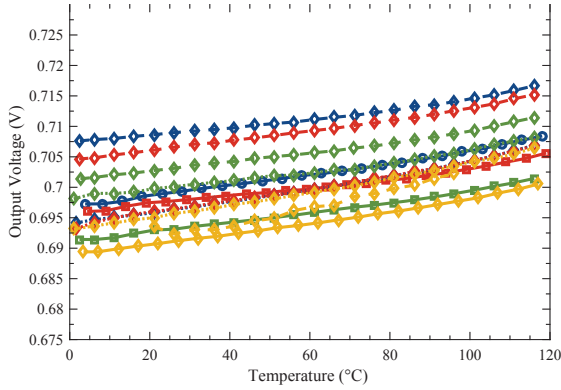


Fig. 7: Measured output voltages as a function of temperature for measured 12 samples with supply voltage set to 1.2 V.

and 5.54 mV respectively. The resulted coefficient of variation σ/μ is 0.79 %. Table I summarizes the characteristics of proposed design with other state-of-the-art designs. Through comparison, it can be seen that the proposed circuit is superior to others nanowatt designs in coefficient of variation and active area occupation while it is comparable to other in power consumption and TC.

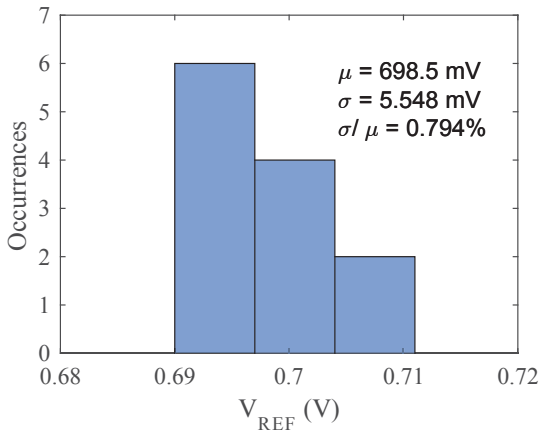


Fig. 8: Distribution of output voltage for 12 measured samples.

IV. CONCLUSION

The ultra low-power voltage reference has been proposed and fabricated in a low-cost CMOS 130 nm technology. The design is entirely implemented by a standard CMOS process and avoids other processes or BJT transistors which result in improved coefficients of variation among nanowatt designs. The design also avoids incorporating any resistor or operational amplifier which leads to extremely small size of 0.007 mm² and it draws only 230 nA from the 1.2 V supply source.

REFERENCES

- [1] R. Sarpeshkar, *Ultra Low Power Bioelectronics : Fundamentals, Biomedical Applications, and Bio-inspired Systems*. Cambridge, UK: Cambridge University Press, 2010.
- [2] K. N. Leung and P. K. T. Mok, "A sub-1-V 15-ppm/ $^{\circ}C$ CMOS Bandgap voltage reference without requiring low threshold voltage device," *IEEE Journal of Solid-State Circuits*, vol. 37, no. 4, pp. 526–530, Apr 2002.
- [3] C. M. Andreou, S. Koudounas, and J. Georgiou, "A Novel Wide-Temperature-Range, 3.9 ppm/ $^{\circ}C$ CMOS Bandgap Reference Circuit," *IEEE Journal of Solid-State Circuits*, vol. 47, no. 2, pp. 574–581, Feb 2012.
- [4] L. Magnelli, F. Crupi, P. Corsonello, C. Pace, and G. Iannaccone, "A 2.6 nW, 0.45 V Temperature-Compensated Subthreshold CMOS Voltage Reference," *IEEE Journal of Solid-State Circuits*, vol. 46, no. 2, pp. 465–474, Feb 2011.
- [5] Y. Osaki, T. Hirose, N. Kuroki, and M. Numa, "1.2-V Supply, 100-nW, 1.09-V Bandgap and 0.7-V Supply, 52.5-nW, 0.55-V Subbandgap Reference Circuits for Nanowatt CMOS LSIs," *IEEE Journal of Solid-State Circuits*, vol. 48, no. 6, pp. 1530–1538, June 2013.
- [6] A. J. Annema, "Low-power bandgap references featuring DTMOSTs," in *Solid-State Circuits Conference, 1998. ESSCIRC '98. Proceedings of the 24th European*, Sept 1998, pp. 116–119.
- [7] K. Ueno, T. Hirose, T. Asai, and Y. Amemiya, "A 300 nW, 15 ppm/ $^{\circ}C$, 20 ppm/V Cmos Voltage Reference Circuit Consisting of Subthreshold MOSFETs," *IEEE Journal of Solid-State Circuits*, vol. 44, no. 7, pp. 2047–2054, July 2009.
- [8] A. J. Annema and G. Goksun, "A 0.0025mm² bandgap voltage reference for 1.1V supply in standard 0.16 μm cmos," in *2012 IEEE International Solid-State Circuits Conference*, Feb 2012, pp. 364–366.
- [9] G. D. Vita and G. Iannaccone, "A sub-1-V, 10 ppm/ $^{\circ}C$, Nanopower Voltage Reference Generator," *IEEE Journal of Solid-State Circuits*, vol. 42, no. 7, pp. 1536–1542, July 2007.